EPD INTERNATIONAL EPD SYSTEM

SOUTHEAST ASIA EPD INTERNATIONAL EPD SYSTEM

ECO PLATFORM EPD VERIFIED

Environmental Product Declaration of multiple products based on the average results of the product group

In accordance with ISO 14025:2006 and EN 15804:2012+A2:2019/AC:2021 for:

Hot Rolled Plate

From

GRP SHAPING TOMORROW

PT Gunung Raja Paksi Tbk

Programme
The International EPD® System, www.environdec.com

EPD registered through the fully aligned regional hub
EPD Southeast Asia, https://www.epd-southeastasia.com/

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General information

Programme information

Programme	**The International EPD ® System** EPD registered through the fully aligned regional hub: EPD Southeast Asia	
Address:	**EPD International AB** Box 210 60, SE-100 31 Stockholm, Sweden **EPD Southeast Asia** Kencana Tower Level M, Business Park Kebon Jeruk Jl. Raya Meruya Ilir No. 89, Jakarta Barat 11620, Indonesia	**Website** www.environdec.com www.epd-southeastasia.com **Email** info@environdec.com

ISO standard ISO 21930 and CEN standard EN 15804 serves as the core Product Category Rules (PCR)
Product category rules (PCR): PCR 2019:14 Construction products (EN 15804+A2), Version 2.0.1, 2025-06-05 (valid until 2030-04-07)
PCR review was conducted by: The Technical Committee of the International EPD® System. **Review chair:** Rob Rouwette The review panel may be contacted via the Secretariat www.environdec.com/contact.
Life Cycle Assessment (LCA): LCA Accountability: PT Life Cycle Indonesia Corresponding practitioner: Jessica Hanafi, Ph.D., Gloria FJ Kartikasari, Ruth Medyani S, Vivian.
Independent third-party verification of the declaration and data, according to ISO 14025:2006: ☑ Individual EPD verification without a pre-verified LCA/EPD tool
Third party verifier: Claudia Peña, Director of PINDA LCT SpA **Approved by:** The International EPD® System Technical Committee, supported by the Secretariat
Procedure for follow-up of data during EPD validity involves third party verifier: ☐ Yes ☑ No

The EPD owner has the sole ownership, liability, and responsibility for the EPD.

EPDs within the same product category but published in different EPD programmes, may not be comparable. For two EPDs to be comparable, they shall be based on the same PCR (including the same first-digit version number) or be based on fully aligned PCRs or versions of PCRs; cover products with identical functions, technical performances and use (e.g. identical declared/functional units); have identical scope in terms of included life-cycle stages (unless the excluded life-cycle stage is demonstrated to be insignificant); apply identical impact assessment methods (including the same version of characterisation factors); and be valid at the time of comparison.

Information about EPD owner

EPD Owner
PT Gunung Raja Paksi Tbk

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Contact person:
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Description of the organisation

Established in 1970 in Medan, North Sumatra, the company with the name of PT Gunung Naga Mas, started the business by producing hot steel, gradually producing beams and steel sheets. In 1991, PT Gunung Naga Mas transitioned to PT Gunung Raja Paksi (GRP). GRP is located in Cikarang Barat, West Java Province, Indonesia, covering more than 200 hectares. By now, GRP has achieved production capacity of around 2.2 million tonnes per year. In the coming years, GRP will continue to grow and ensure the fulfillment of the need for high quality steel products.

PT Gunung Raja Paksi Tbk (GRP) is one of the largest private steel companies in Indonesia. To become a world-class integrated steel manufacturer, GRP always nurtures a culture of continuous improvement through achievement and advancement in all fields of development. The mission of the company is to ensure stakeholders' and customers' satisfaction through innovation, efficiency, productivity, quality products and services as well as company social responsibility.

GRP continues to show sustainable growth in all areas. Until now, there are more than 4500 qualified employees, who have undergone rigorous training, helping Gunung Steel Group (GSG) achieve corporate success in Indonesia. The company is committed to always operating beyond compliance, including on the aspect of the environment. Therefore, GRP is very concerned about the quality of products and its management systems. The ability of GRP has

been proven by international market customers in Asia, South Asia, Australia, Africa, Europe, Middle East, and America.

Product-related or management system-related certifications

This is evident from the many international certifications obtained such as

- ACRS (Australasian Certification Reinforcing and Structural Steel) 3678 certification no 171202, ACRS 3679 certification no 171203, and ACRS ASNZS 3679 certification no T 1011
- American Standard of Building Fabricators (AWS, ASTM) – AISC American Institute of Steel Construction)
- Division 3 STEEL STRUCTURE FABRICATION CSA W 47.1 – CWB
- ISO 17025:2017
- ISO 3834 (Welding Management System) WTIA/Weld Australia certification no AU 030 - Rev.0
- License for Fabrication Activity in Los Angeles – LADBS
- LR (Lloyd's Register) QA ISO 9001:2015 no 10115920
- Membership of Climate Action CO2 Worldsteel Association

Details of GRP's commitment to sustainable development can be found in the company's sustainability report.

Name and location of production site

Jl. Perjuangan No.15, RT.004/RW.006, Sukadanau, Kec. Cikarang Barat, Bekasi 17530, West Java, Indonesia

Product information

Product name

Hot Rolled Plate

Product identification

AS/NZS 3678:2016, ACRS Certificate No.171202

Product description

Hot rolled plate is made of steels which where deformed in the continuous/semi-continuous type hot rolling mill with AGC facilities using X-ray or equivalent thickness gauge to monitor and control variation of thickness along the length to gain a desirable dimension of a steel plate.

This study covers several types of hot rolled plate products, namely Structural Mild Steel and High Strength Steel, as presented in the table below. Any differences in the composition of these steel grades are generally considered insignificant to the overall results, with an average variation of no more than 10% from cradle to gate. Therefore, the environmental performance results presented in this report represent the average performance of the product group

Table 1. Steel Type

Steel Type	General Description	Typical Uses	Specification Standard
Structural Mild Steel	Hot rolled Structural Mild Steel with nominal 250 MPa yield strength. Including additional testing requirements for impact with different range of temperature.	• Structural steel for general construction • Structural Steel Work for fabrication and erection • Bridge design • Buildings • Storage Tanks • Weather Resistant Steel Structure	1. AS/NZS 3678 : 2016 2. AS/NZS 1365 : 1996 (reconfirmed 2016)
High Strength Steel	Hot rolled High Strength Steel with nominal 350 MPa yield strength. Including additional testing requirements for impact with different range of temperature.		
Weather Resistant Steel	Hot Rolled Weather Resistant Steel with nominal 350 Mpa Yield strength and Weathering steel is therefore classed as being a 'self–protecting' steel. Including additional testing requirements for impact of different range of temperature		

Plate Size

Maximum width is 3 000 mm and length is 12 000 mm which GRP capable to supply. However, other dimensions (custom) may be ordered by agreement.

Table 2. Standard GRP Plate Size

Standard Plate Size	Width (mm)	1 200	1 219	1 500	1 520	1 800	1 830
	Length (mm)	2 400	2 438	6 000	6 096	6 000	6 096

Fabricating Performance

Fabricating performance is an indicator of the steel plate to show the level of facility and suitability of the material to perform into other products such as welding or cold forming. GRP Product is applicable to have bending and welding method with succesfull rate ≥ 8

Table 3. Fabricating Performance

1 = Limited	10 = Excellent
Method	**Rating**
Bending	8
Welding	9

Specification Standard

The standar specification is based on AS/NZS with the following grade of 200; 250; 300; 350 and 400. Additional thickness test i.e., L15; L20; L40 are also shown in table 4, table 5 and table 6.

Table 4. Mechanical Properties.

Specification & Grades	Applicable Thickness in GRP (mm)	Thickness Range (mm)	Tensile Test (Transversal)		
			YS (N/mm²)	TS (N/mm²)	Elongation (%)
			Min	Min	Min
AS/NZS 3678 :2016 Gr. 200, 200 L0; 200 L15; 200 L20	≥5 - ≤100	≤ 8	200	300	22[B]
		> 8 ≤ 12	200	300	22[B]
		> 12 ≤ 20	-	300	22[B]
		> 20 ≤ 50	-	300	22[B]
		> 50 ≤ 80	-	300	22[B]
		> 80 ≤ 150	-	300	22[B]
AS/NZS 3678 :2016 Gr. 250, 200 L0; 250 L15; 250 L20	≥5 - ≤100	≤ 8	280	410[A]	22[B]
		> 8 ≤ 12	260	410	22[B]
		> 12 ≤ 20	250	410	22[B]
		> 20 ≤ 50	250	410	22[B]
		> 50 ≤ 80	240	410	22[B]
		> 80 ≤ 150	230	410	22[B]

Specification & Grades	Applicable Thickness in GRP (mm)	Thickness Range (mm)	Tensile Test (Transversal)		
			YS (N/mm²)	TS (N/mm²)	Elongation (%)
			Min	Min	Min
AS/NZS 3678 :2016 Gr. 300, 300 L0; 300 L15; 300 L20; 300 L40	≥8 - ≤100	≤ 8	320	430	21[b]
		> 8 ≤ 12	310	430	21[b]
		> 12 ≤ 20	300	430	21[b]
		> 20 ≤ 50	280	430	21[b]
		> 50 ≤ 80	270	430	21[b]
		> 80 ≤ 150	260	430	21[b]
AS/NZS 3678 :2016 Gr. 350, 350 L0 350 L15; 350 L20; 350 L40	≥8 - ≤100	> 8 ≤ 12	360	450	20[b]
		> 20 ≤ 32	340	450	20[b]
		> 32 ≤ 80	340	450	20[b]
		> 80 ≤ 100	330	450	20[b]
AS/NZS 3678 :2016 Gr. 350 WR L0, 350 WR L15; 350 WR L20;	≥8 - ≤60	> 8 ≤ 12	360	450	20[b]
		> 20 ≤ 32	340	450	20[b]
		> 32 ≤ 60	340	450	20[b]
		> 80 ≤ 100	330	450	20[b]
AS/NZS 3678 :2016 Gr. 400, 400 L15; 400 L20	≥8 - ≤32	≤ 8	400	480	18[b]
		> 8 ≤ 12	400	480	18[b]
		> 12 ≤ 20	380	480	18[b]
		> 20 ≤ 50	360	480	18[b]
		> 50 ≤ 80	360	480	18[b]
AS/NZS 3678 :2016 Gr. 300 S0	≥8 - ≤80	≤ 8	320	430	25
		> 8 ≤ 12	310	430	25
		> 12 ≤ 20	300	430	25
		> 20 ≤ 50	280	430	25
		> 50 ≤ 80	270	430	25
		> 80 ≤ 150	260	430	25

Specification & Grades	Applicable Thickness in GRP (mm)	Thickness Range (mm)	Tensile Test (Transversal)		
			YS (N/mm²)	TS (N/mm²)	Elongation (%)
			Min	Min	Min
AS/NZS 3678 :2016 Gr. 350 S0	≥8 - ≤80	> 8 ≤ 12	360	450	25
		> 20 ≤ 32	340	450	25
		> 32 ≤ 80	340	450	25
		> 80 ≤ 100	330	450	25
AS/NZS 3678 :2016 Gr. 350 WR S0	≥20 - ≤40	> 8 ≤ 12	360	450	25
		> 20 ≤ 32	340	450	25
		> 32 ≤ 80	340	450	25
		> 80 ≤ 100	330	450	25

Remarks:

[A] Minimum tensile strength (TS) is not applicable for material with thickness < 6 mm.
[B] For cross sectional test piece area > 1 000 mm², minimum elongation decreased by 2%.

Table 5. Impact Properties.

Specification	Grades	Temp. (°C)	Minimum Absorbed Energy (J) [B]
			Avg.
AS/NZS 3678 : 2016	Gr. 250/300/350/350 WR L0	0	27
	Gr. 200/250/300/350/ 350WR/400 (L15)	-15	27
	Gr. 200/250/300/350/ 350WR/400 (L20)	-20	27
	Gr. 300/350 (L40)[A]	-40	27
	Gr. 300/350/350WR (S0)	0	70

[A] Supplied in normalized condition up to 100 mm thickness to guarantee impact value
[B] Average Value of 3 Sample.

Table 6. Through Thickness (Z-Test) Properties

Through Thickness (AS/NZS 3678:2016)				
Grade	Certified Thickness in GRP (mm)	Z 15	Z 25	Z 35
200	12-100	✓	✓	✓
250	12-100	✓	✓	✓
300	12-100	✓	✓	✓
350	12-60	✓	✓	✓
400	12-32	✓	-	✓

Through Thickness Test (Z-Test), the product is capable from thickness 15 mm and above.

UN CPC code

41211 -Flat-Rolled Products of Non-Alloy Steel, Not Further Worked than Hot-Rolled, of a Width of 600 mm or More

ANZSIC code

2110 - Iron Smelting and Steel Manufacturing

Geographical Scope

Manufactured in Indonesia, distributed to New Zealand and Singapore

LCA information

Declared Unit

1 000 kg of
hot rolled plate

Reference service life

Not applicable

Time representativeness

specific data for the manufacturing collected from 2024-01-01 to 2024-12-31. The 10-year age requirement for generic data has been met.

Database(s) and LCA software used

generic data for upstream and downstream processes use Ecoinvent 3.10 database and modelled by using SimaPro Developer software version 10.2.0.1. No datasets older than 10 years were used.

Description of system boundaries

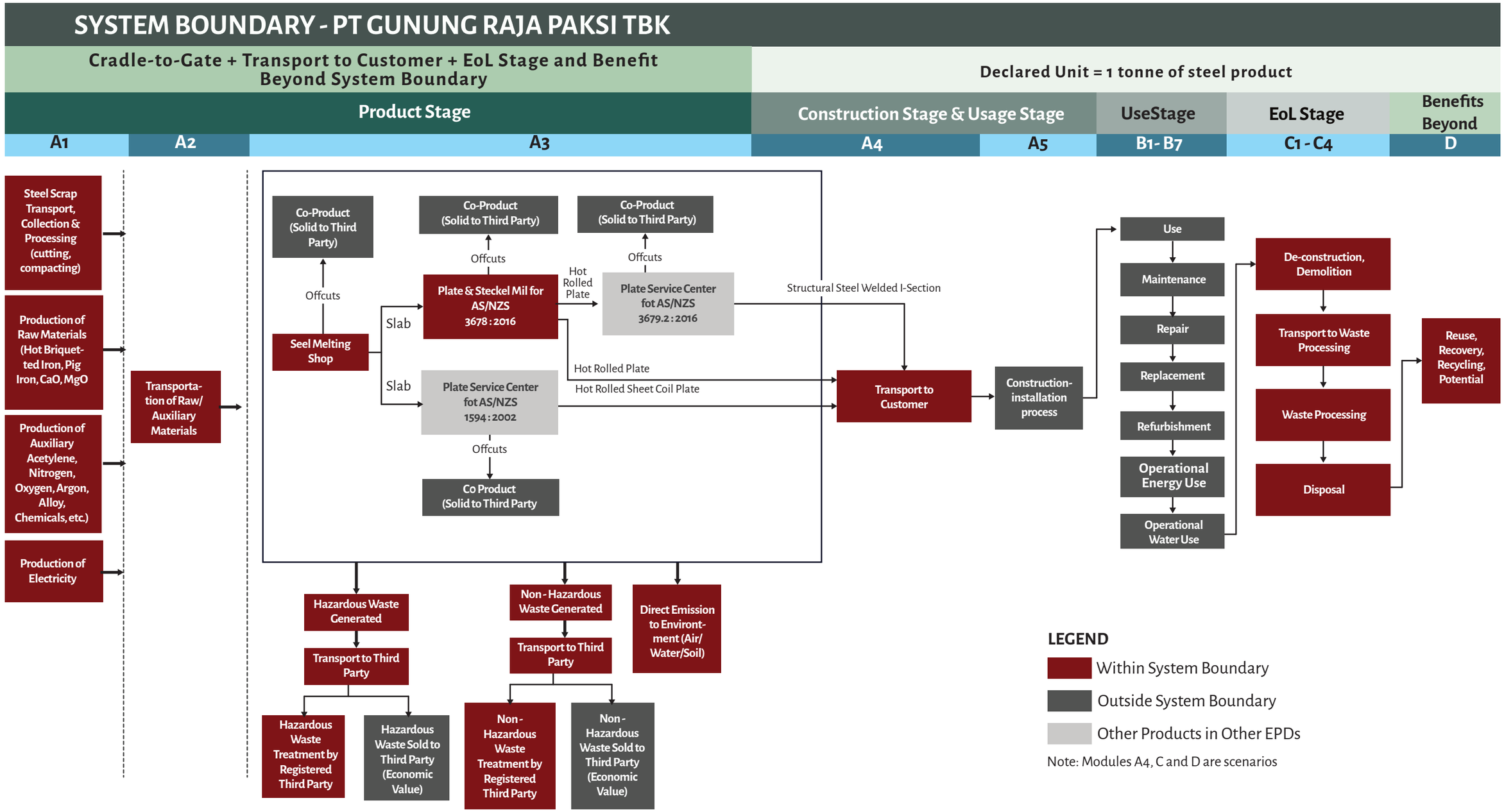
The system boundary was chosen based on the goal and scope of the study and in accordance with EN 15804:2012+A2:2019, i.e. “cradle-to-gate” with options, modules C1–C4 (the end-of-life stages), module D (benefits and loads beyond the system boundary), and with optional modules A4. Modules A5 and B1-B7 have not been included due to the inability to predict how the material will be used in the construction process and use stage. The processes below are included in the product system to be studied:

1. **Upstream (A1-A2)**
   a. Steel Scrap collection & processing
   b. Production of raw materials (Hot Briquetted Iron, Pig Iron, CaO, MgO)
   c. Production of auxiliary materials in the form of solid, liquid or gas (e.g. Alloy, Chemicals, Electrode, Acetylene, Argon, Nitrogen, Oxygen, LNG, etc.)
   d. Production of electricity from electricity mix in Indonesia from Ecoinvent Database
   e. Transportation of raw/auxiliary materials from the supplier to manufacturing plant

2. **Core (A3)**
   c. Steel Melting Shop (SMS): Electric Arc Furnace, Ladle Furnace, Tundish, Continuous Casting Machine, Cutting, Marking, Scarfing
   d. Plate & Steckel Mill (PSM): Reheating Furnace, Descale (Primary & Secondary), Roughing Mill, Steckel Furnace, Steckel Mill, Finishing Mill, Laminar Cooling, Up Coiler, Hot Leveler, Cooling Bed, Normalizing, AUT Machine, Shot Blasting & Painting, Inspection & Repair, Cooling, Equalizer, Cutting Process (Hot & Cold/Cold), Marking, Storage
   e. Clarifier (vessel)
   f. Dust collector & blower
   g. Waste treatment (slag, skull, scrap, welding wire, electrode, grinding disc, etc.)
   h. Hazardous waste generated and waste treatment by the registered third party
   i. Non-hazardous waste generated that is sold to third parties
   j. Direct emissions to the environment

3. **Downstream (A4, C1-C4, D)**
   d. Transport to the building/construction site
   e. Deconstruction & Demolition
   f. Transport to waste processing facilities
   g. Waste processing of the scrap steel
   h. Disposal
   i. Reuse/Recovery/Recycling at the end of life of the products

System diagram



PT Gunung Raja Paksi Tbk
GRP SHAPING TOMORROW

More information

Relevant website for more information regarding the manufacturing process: **www.gunungrajapaksi.com**

Key Assumptions and Limitations

- Production process of materials in upstream process taken from Ecoinvent database reflects average or generic production and therefore does not correspond to actual suppliers.
- Any differences in composition of steel grades within the grade groups is generally considered insignificant compared to the overall results. A sensitivity analysis was conducted, showing that the average overall variation is insignificant. Hence, the impact assessment was grouped.
- Land use change emissions in module A3 were considered immaterial. The plant is in an industrial zone which was established in 1990 (more than 30 years ago).
- The water consumption was counted from the amount of makeup water to compensate the losses due to water evaporation.

Cut-off rules

In case of insufficient input data or data gaps for a unit process, the cut-off criteria shall be 1 % of renewable and non-renewable primary energy usage and 1 % of the total mass input of that unit process. The total of neglected input flows per module, e.g., per module A1-A3, A4, C1-C4 and module D shall be a maximum of 5 % of energy usage and mass. In this study, all data in the product system is included. If there is missing specific data, generic data from the database or literature was used.

Data Quality

- Time related coverage: specific data were collected from 2024-01-01 to 2024-12-31 and generic data are representative of the year 2024.
- Geographic coverage: specific data were collected from the area under study, i.e., West Java, Indonesia. Electricity production as a key input is sourced from generic data of Indonesia Grid available in database. Generic data were collected from global average data.
- Technological coverage: specific data were collected from current steel making process under study. There is no specific data for upstream (module A1-A2) and downstream process (module A4, C1-C4), therefore generic data from the global average was used with similar technology aspects to describe the process under study.

Data quality for both specific and generic data were sufficient to conduct life cycle assessment in accordance with the defined goal and scope.

Allocation

Economic allocation was applied in accordance with EN 15804:2012+A2:2019. Allocation was applied to allocate the main product and the steel scrap coming out the manufacturing i.e., cutting scale and mill scale. For the end-of-life of waste generated in the manufacturing process (e.g., slag), polluter pays principle are applied for each type of waste. This means that GRP will carry the full environmental impact until the end-of-waste state is reached.

Multi-input allocation is relevant for any material sent to landfill, i.e. steel scrap and also waste generated in the manufacturing. The emissions to air and soil (from leaching) are determined based on the physical/chemical composition of the inputs or physical properties of the material going to

landfill. As a result, steel products or other waste generated (paint container, used used spray paint containers, penetrant, etc.) that end up in landfill only contribute to those emissions that are likely to occur based on the input material. Overhead processes associated with landfill (e.g. energy used in equipment for managing the landfill site) are attributed to waste flows based on their mass. Mass allocation is considered a reasonable estimation for attributing overhead processes to various waste flows. This allocation is applied as well for any material sent to incineration, i.e. contaminated gloves and rags

In this study, the closed-loop process is applied between modules A-C and module D. When the scrap is used in the manufacture of a new product, there is an allocation (or debit) associated with the scrap input. Meanwhile the recovered steel scrap for recycling is allocated a credit (or benefit) associated with the avoided impacts of the virgin material. If the amount of recovered steel scrap for recycling is less than what the product system requires/steel scrap needed in the manufacture, then the environmental burdens associated with meeting the raw material demand are included in this closed-loop model. If, however, the amount of recovered steel scrap for recycling is larger than what the product system requires/steel scrap needed in the manufacture, then the product system receives a net credit, equivalent to the net amount of virgin material avoided.

The recovered steel scrap that is not looped back to the manufacture (leaving product system that have passed the end-of-waste state), goes to module D, except those which have been allocated as co-product. The end-of-waste state of the steel scrap is reached when the steel scrap is processed in the waste processing (Module C3). The steel scrap is sorted and pressed into blocks and ready to be used for other specific purposes. After the point of end-of-waste, the downstream emissions related to transportation process from recycler to manufacture is attributed to the processing unit that uses the secondary material.

The impacts assigned to the credit or burden that comes from module D are calculated by adding impact connected to secondary steel production from EAF plant (beyond system boundary) and subtracting the impacts resulting from primary steel production at BOS plant. The difference between 100% primary steel production (BOS plant) and 100% secondary steel production (EAF plant) is the result of the module D. The calculation is following worldsteel methodology of steel scrap. The benefit beyond system boundary (module D) is a credit estimation resulted from the system because in real-life there is a trans-continent boundary of the market in Australia/New Zealand and producers in Indonesia which do not share the recycled material market. The assessed products are exported to Australia and New Zealand. Therefore, the recovery rate for recycling is adjusted to the rate in each country, i.e., 90% in New Zealand and Australia. Around 10% of the steel scrap is considered as material losses that will go to another disposal scenario to landfill.

LCA Scenarios and Additional Technical Information

- Electricity grid was based on the modified Ecoinvent database for Indonesia electricity network. This modification followed a conventional residual mix approach. The resulting climate change impact (GWP-GHG) for the grid mix is 1.31 kg CO_2 eq./kWh.
- Transportation using truck in Indonesia use EURO3 to represent the current condition. Meanwhile in Singapore and New Zealand, EURO5 is used as a standard emission.
- Transport distance was calculated by Google Maps from GRP to Port of Tanjung Priok Indonesia (47 km) and calculated by Sea Route & Distance Calculator from Port of Tanjung Priok to destination port.
- Amount of diesel used for demolition process was modelled using Ecoinvent database (Waste reinforcement steel {RoW}| treatment of waste reinforcement steel, recycling | Cut-off, U) for global data, i.e., 0.626 MJ diesel/kg steel.
- Amount of diesel and electricity consumption for waste processing was modelled using Ecoinvent database for global data on

sorting and pressing iron scrap, i.e., 0.1 MJ diesel/kg steel and 0.01 kWh/kg steel.

- Average recycling rate for steel is 90% in New Zealand according to New Zealand Steel (2021). Meanwhile in Singapore it is 99% in Singapore according to The National Environment Agency (2021). The remaining steel scrap was considered as material losses to the landfill..
- GRP uses external scrap in its steel production. Net scrap was calculated by excluding the amount of internal scrap (home scrap). The potential environmental benefit calculated for the end-of-life stage (Module D) was based on the net amount of scrap left in the system in accordance with "value of scrap" worldsteel methodology.
- Electricity was modelled using Ecoinvent database for Singapore and New Zealand
- The characterization method follows EN 15804:2012+A2:2019 with EF 3.1 (July 2022). A supplementary mandatory indicator for climate impact (GWP-GHG) with IPCC 2021 GWP100 (AR6) method, is reported as well in accordance with the PCR.

Modules declared, geographical scope, share of specific data (in GWP-GHG indicator) and data variation

	Product stage			Con-struction process stage		Use stage							End of life stage				Resource recovery stage
	Raw material supply	Transport	Manufacturing	Transport	Construction installation	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal	Reuse-Recovery-Recycling-potential
Module	A1	A2	A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
Modules declared	X	X	X	X	ND	ND	ND	ND	ND	ND	ND	ND	X	X	X	X	X
Geography	GLO	GLO	ID	SG, NZ	-	-	-	-	--	-	-	-	SG, NZ	SG, NZ	SG, NZ	SG, NZ	GLO
Primary data used	65%			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation – products	<10%			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation – sites	Not Relevant			-	-	-	-	-	-	-	-	-	-	-	-	-	-

The share of primary data is calculated based on GWP-GHG results. It is a simplified indicator for data quality that supports the use of more primary data, to increase the representativeness of and comparability between EPDs. Note that the indicator does not capture all relevant aspects of data quality and is not comparable across product categories.

The declaration of data sources, reference years, data categories, and share of primary data is summarized in the table below. The processes listed in the table represent those that contribute more than 10% to the GWP-GHG results of modules A1–A3.

Process	Source Type	Source	Reference Year	Data Category	Share of Primary Data (GWP A1-A3)
Energy For production (Electricity & Natural Gas)	Database, Collected Data	Ecoinvent 3.10, EPD Owner	2024	Primary Data, Secondary Data	64%
Transportation & Distribution activities	Database	Ecoinvent 3.10	2024	Primary Data	2%

Content declaration

Hot rolled steel plate AS/NZS 3678:2016 manufactured by GRP is made of low alloy steels with pig iron and approximately 77% scrap-based material. GRP followed the chemical range of AS/NZS 3678:2016 as per spec, therefore, our typical chemical composition can be seen below .

Product content / declared unit	Mass, kg	Post-consumer recycled material, mass-% of product	Biogenic material, mass-% of product	Biogenic material, kg C/declared unit
Iron (virgin sources)	234	0.00	0	0
Recycled Material (pre- and post-consumer)	766	76.60	0	0
Total	1000	76.60	0	0

Hazardous Substances
The product does not contain SVHC substances above 0.1%

Chemical composition, %	AS/NZS 3678:2016 Gr. 200, 200 L0; 200 L15; 200 L20	AS/NZS 3678:2016 Gr. 250, 250 L0; 250 L15; 250 L20	AS/NZS 3678:2016 Gr. 300, 300 L0; 300 L15; 300 L20; 300 L40; 300 S0	AS/NZS 3678:2016 Gr. 350, 350 L0; 350 L15; 350 L20; 350 L40; 350 S0	AS/NZS 3678:2016 Gr. 350 WR; 350 WR S0	AS/NZS 3678:2016 Gr. 400, 400 L15; 400 L20
Iron (Fe)	97-99	97-99	97-99	97-99	97-99	97-99
Carbon (C)	0.09-0.12	0.16-0.20	0.17-0.20	0.17-0.20	0.11-0.14	0.17-0.20
Silica (Si)	0.10-0.15	0.20-0.30	0.20-0.30	0.20-0.30	0.25-0.35	0.20-0.35
Manganese (Mn)	0.3-0.4	0.40-0.90	0.8-1.20	1.20-1.40	1.15-1.40	1.20-1.45
Phosphorus (P)	≤0.015	≤0.015	≤0.015	≤0.015	≤0.015	≤0.015
Sulphur (S)	≤0.010	≤0.010	≤0.010	≤0.010	≤0.010	≤0.010
Chrome (Cr)	-	-	-	-	0.40-0.50	-

Packaging materials
No packaging used for the products

Environmental performance

The characterization method follows EN 15804:2012+A2:2019 with EF 3.1 (July 2022). A supplementary mandatory indicator for climate impact (GWP-GHG) with IPCC 2021 GWP100 (AR6) method, is reported as well in accordance with the PCR.

The potential environmental impact indicators along with the characterization method are described in the table below.

The results of the end-of-life stage (modules C1-C4) should be considered when using the results of the product stage (modules A1-A3).

Impact Indicator	Abbreviation	Unit	Characterization Method
Potential Environmental Impact Indicators, in accordance with EN 15804:2012+A2:2019			
Global Warming Potential - total	GWP-total	kg CO_2 eq.	IPCC 2021
Global Warming Potential - fossil fuels	GWP-fossil	kg CO_2 eq.	IPCC 2021
Global Warming Potential - biogenic	GWP-biogenic	kg CO_2 eq.	IPCC 2021
Global Warming Potential - land use and land-use change	GWP-luluc	kg CO_2 eq.	IPCC 2021
Depletion potential of the stratospheric ozone layer	ODP	kg CFC11 eq.	WMO 2014
Acidification potential	AP	mol H^+ eq.	Accumulated Exceedance
Eutrophication potential - freshwater	EP-freshwater	kg P eq.	EURTREND model (ReCiPe)
Eutrophication potential - marine	EP-marine	kg N eq.	EURTREND model (ReCiPe)
Eutrophication potential - terrestrial	EP-terrestrial	mol N eq.	Accumulated Exceedance
Formation potential of tropospheric ozone	POCP	kg NMVOC eq.	LOTUS-EUROS
Abiotic depletion potential for non-fossil resources	ADP-minerals & metals	kg Sb eq.	CML 2002a
Abiotic depletion potential for fossil resources	ADP-fossil	MJ, net calorific value	CML 2002a
Water use deprivation potential	WDP	m^3 world eq. deprived	AWARE

Impact Indicator	Abbreviation	Unit	Characterization Method
Additional Environmental Impact Indicators, in accordance with EN 15804:2012+A2:2019			
Particulate Matter emissions	PM	Disease incidence	SETAC-UNEP, Fantke et al. 2016
Ionizing radiation - human health	IRP	kBq U235 eq.	Human Health Effect model
Eco-toxicity - freshwater	ETP-fw	CTUe	USEtox
Human toxicity - cancer effects	HTP-c	CTUh	USEtox
Human toxicity - non-cancer effects	HTP-nc	CTUh	USEtox
Land use related impacts / soil quality	SQP	dimensionless	Soil quality index (LANCA®)
Climate Impact, in accordance with PCR 2019-14 Construction products v2.01			
Climate Impact	GWP-GHG	kg CO2 eq.	IPCC 2021 (AR6)
Resource Use Parameters			
Use of renewable primary energy excluding renewable primary energy resources used as raw materials	PERE	MJ	N/A
Use of renewable primary energy resources used as raw materials	PERM	MJ	N/A
Total use of renewable primary energy resources	PERT	MJ	N/A
Resource Use Parameters			
Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials	PENRE	MJ	N/A
Use of non-renewable primary energy resources used as raw materials	PENRM	MJ	N/A
Total use of non-renewable primary energy resources	PENRT	MJ	N/A
Use of secondary material	SM	kg	N/A
Use of renewable secondary fuels	RSF	MJ	N/A
Use of non-renewable secondary fuels	NRSF	MJ	N/A
Use of net fresh water	FW	m^3	N/A

Impact Indicator	Abbreviation	Unit	Characterization Method
Waste Categories and Output Flows			
Hazardous waste disposed	HWD	kg	N/A
Non-hazardous waste disposed	NHWD	kg	N/A
Radioactive waste disposed	RWD	kg	N/A
Components for reuse	CRU	kg	N/A
Materials for recycling	MFR	kg	N/A
Materials for energy recovery	MER	kg	N/A
Exported electrical energy	EEE	MJ	N/A
Exported thermal energy	EET	MJ	N/A

Potential environmental impact – mandatory indicators according to EN 15804

The LCIA results are relative expressions and do not predict impacts on category endpoints, the exceeding of thresholds, safety margins or risks.

Results for 1 tonne of Hot Rolled Plate (AS/NZS 3678:2016)								
Impact Indicator	Unit	Total A1-A3	A4	C1	C2	C3	C4	D
GWP-total	kg CO_2 eq.	1.8E+03	9.3E+01	6.3E+01	4.9E+00	1.4E+01	9.8E-01	7.0E+00
GWP-fossil	kg CO_2 eq.	1.7E+03	9.3E+01	6.3E+01	4.9E+00	1.4E+01	9.8E-01	7.0E+00
GWP-biogenic	kg CO_2 eq.	6.8E-01	1.4E-02	5.6E-03	1.8E-04	2.0E-03	1.8E-04	-5.7E-03
GWP-luluc	kg CO_2 eq.	5.4E-01	4.8E-02	5.4E-03	1.2E-04	9.1E-03	3.2E-04	2.0E-03
ODP	kg CFC11 eq.	1.2E-05	1.4E-06	9.6E-07	6.7E-08	2.1E-07	2.3E-08	1.1E-08
AP	mol H^+ eq.	9.5E+00	2.6E+00	5.7E-01	1.1E-02	1.1E-01	5.3E-03	2.7E-02
EP-freshwater	kg P eq.	2.3E-01	3.9E-04	2.2E-04	3.5E-06	1.3E-04	6.2E-06	-5.3E-04
EP-marine	kg N eq.	2.0E+00	6.4E-01	2.6E-01	3.9E-03	4.8E-02	2.0E-03	4.0E-03
EP-terrestrial	mol N eq.	2.2E+01	7.1E+00	2.9E+00	4.3E-02	5.2E-01	2.2E-02	6.7E-02
POCP	kg NMVOC eq.	6.6E+00	1.9E+00	8.6E-01	1.8E-02	1.6E-01	7.9E-03	1.9E-02
ADP-minerals & metals[1]	kg Sb eq.	1.4E-03	1.0E-04	2.2E-05	1.8E-07	5.9E-07	9.6E-07	8.0E-05
ADP-fossil	MJ, net calorific value	2.1E+04	1.2E+03	8.2E+02	6.6E+01	1.8E+02	2.0E+01	6.2E+01
WDP	m^3 world eq. deprived	2.4E+02	2.8E+00	1.8E+00	2.7E-02	-1.1E+00	6.5E-01	6.7E+00

[1] The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator

Potential environmental impact – additional environmental information according to EN 15804:2012+A2:2019

Results for 1 tonne of Hot Rolled Plate (AS/NZS 3678:2016)								
Impact Indicator	Unit	Total A1-A3	A4	C1	C2	C3	C4	D
PM	Disease incidence	1.4E-04	3.1E-06	1.6E-05	3.2E-07	2.8E-06	1.2E-07	4.7E-07
IRP	kBq U235 eq.	8.7E+00	2.4E-01	1.4E-01	4.6E-03	1.8E-02	4.0E-03	5.2E-02
ETP-fw	CTUe	1.5E+04	2.1E+02	1.2E+02	3.1E+00	1.8E+01	2.3E+00	2.4E+02
HTP-c	CTUh	3.2E-05	4.0E-07	2.5E-07	3.7E-10	1.2E-09	2.8E-09	8.7E-07
HTP-nc	CTUh	4.7E-05	3.1E-07	1.0E-07	3.3E-08	3.5E-08	5.1E-09	-4.3E-07
SQP	dimensionless	2.1E+03	1.3E+02	5.8E+01	9.3E-02	7.8E-01	2.9E+01	2.3E+01

Climate impact (GWP-GHG) – according to PCR

Results for 1 tonne of Hot Rolled Plate (AS/NZS 3678:2016)								
Impact Indicator	Unit	Total A1-A3	A4	C1	C2	C3	C4	D
GWP-GHG	kg CO_2 eq.	1.8E+03	9.3E+01	6.3E+01	4.9E+00	1.4E+01	9.8E-01	7.0E+00

Resource use

Results for 1 tonne of Hot Rolled Plate (AS/NZS 3678:2016)								
Parameter	Unit	Total A1-A3	A4	C1	C2	C3	C4	D
PERE	MJ	4.6E+02	9.2E+00	5.0E+00	1.0E-01	7.1E+01	1.5E-01	-9.8E+00
PERM	MJ	0	0	0	0	0	0	0
PERT	MJ	4.6E+02	9.2E+00	5.0E+00	1.0E-01	7.1E+01	1.5E-01	-9.8E+00
PENRE	MJ	2.1E+04	1.2E+03	8.2E+02	6.6E+01	1.8E+02	2.0E+01	6.2E+01
PENRM	MJ	0	0	0	0	0	0	0
PENRT	MJ	2.1E+04	1.2E+03	8.2E+02	6.6E+01	1.8E+02	2.0E+01	6.2E+01
SM	kg	1.8E+03	0	0	0	0	0	0
RSF	MJ	0	0	0	0	0	0	0
NRSF	MJ	0	0	0	0	0	0	0
FW	m^3	7.9E+01	5.5E-01	4.1E-01	1.9E-02	1.6E-01	2.5E-02	-2.0E-01

Waste production and output flows

Waste production

Results for 1 tonne of Hot Rolled Plate (AS/NZS 3678:2016)								
Parameter	Unit	Total A1-A3	A4	C1	C2	C3	C4	D
HWD	kg	8.4E-02	0	0	0	0	0	0
NHWD	kg	2.7E-01	0	0	0	0	9.7E+01	0
RWD	kg	1.1E-04	0	0	0	0	0	0

Output flows

Results for 1 tonne of Hot Rolled Plate (AS/NZS 3678:2016)								
Parameter	Unit	Total A1-A3	A4	C1	C2	C3	C4	D
CRU	kg	0	0	0	0	0	0	0
MFR	kg	1.5E+02	0	0	0	9.0E+02	0	0
MER	kg	1.5E+00	0	0	0	0	0	0
EEE	MJ	0	0	0	0	0	0	0
EET	MJ	0	0	0	0	0	0	0

Additional Environmental Information

In compliance with PCR 2.0.1, the following section illustrates theoretical alternative end-of-life scenarios in which the product is 100% landfilled and100% recycled.

Potential environmental impact – mandatory indicators (scenario)

Impact Indicator	Unit	100% Landfill				100% Recycled			
		C1	C2	C3	C4	C1	C2	C3	C4
GWP-total	kg CO_2 eq.	6.3E+01	4.9E+00	-	1.1E+01	6.3E+01	4.9E+00	1.4E+01	-
GWP-fossil	kg CO_2 eq.	6.3E+01	4.9E+00	-	1.1E+01	6.3E+01	4.9E+00	1.4E+01	-
GWP-biogenic	kg CO_2 eq.	5.6E-03	1.8E-04	-	1.9E-03	5.6E-03	1.8E-04	2.0E-03	-
GWP-luluc	kg CO_2 eq.	5.4E-03	1.2E-04	-	3.3E-03	5.4E-03	1.2E-04	9.1E-03	-
ODP	kg CFC11 eq.	9.6E-07	6.7E-08	-	2.5E-07	9.6E-07	6.7E-08	2.1E-07	-
AP	mol H^+ eq.	5.7E-01	1.1E-02	-	5.6E-02	5.7E-01	1.1E-02	1.1E-01	-
EP-freshwater	kg P eq.	2.2E-04	3.5E-06	-	6.5E-05	2.2E-04	3.5E-06	1.3E-04	-
EP-marine	kg N eq.	2.6E-01	3.9E-03	-	2.1E-02	2.6E-01	3.9E-03	4.8E-02	-
EP-terrestrial	mol N eq.	2.9E+00	4.3E-02	-	2.3E-01	2.9E+00	4.3E-02	5.2E-01	-
POCP	kg NMVOC eq.	8.6E-01	1.8E-02	-	8.4E-02	8.6E-01	1.8E-02	1.6E-01	-
ADP-minerals & metals[1]	kg Sb eq.	2.2E-05	1.8E-07	-	9.9E-06	2.2E-05	1.8E-07	5.9E-07	-
ADP-fossil	MJ, net calorific value	8.2E+02	6.6E+01	-	2.2E+02	8.2E+02	6.6E+01	1.8E+02	-
WDP	m^3 world eq. de-prived	1.8E+00	2.7E-02	-	6.7E+00	1.8E+00	2.7E-02	-1.1E+00	-

[1] The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experienced with the indicator

Potential environmental impact – additional environmental information (scenario)

Impact Indicator	Unit	100% Landfill				100% Recycled			
		C1	C2	C3	C4	C1	C2	C3	C4
PM	Disease incidence	1.6E-05	3.2E-07	-	1.3E-06	1.6E-05	3.2E-07	2.8E-06	-
IRP	kBq U235 eq.	1.4E-01	4.6E-03	-	4.2E-02	1.4E-01	4.6E-03	1.8E-02	-
ETP-fw	CTUe	1.2E+02	3.1E+00	-	2.4E+01	1.2E+02	3.1E+00	1.8E+01	-
HTP-c	CTUh	2.5E-07	3.7E-10	-	2.9E-08	2.5E-07	3.7E-10	1.2E-09	-
HTP-nc	CTUh	1.0E-07	3.3E-08	-	5.9E-08	1.0E-07	3.3E-08	3.5E-08	-
SQP	dimensionless	5.8E+01	9.3E-02	-	3.0E+02	5.8E+01	9.3E-02	7.8E-01	-

Interpretation of results

- Module A1-A3 contributes significantly to the impact generated by the whole life cycle.
- The electricity production process is the largest contributor to majority of the potential impacts followed by cast iron production.
- Transport to customer (module A4) makes a relatively small contribution to life cycle emissions for most indicators. However, it is significant for several impact categories, such as, ozone depletion potential, acidification potential, eutrophication potential, and formation potential of tropospheric ozone.
- Activities carried out on the end-of-life of the steel life cycle (module C1-C4) did not have a significant impact on the overall steel life cycle studied.

Additional Social and Economic Information

PT Gunung Raja Paksi Tbk cooperates with Institut Pertanian Bogor (IPB) or Bogor Agricultural University, are managing 7 green areas at GRP covering the employee mess area and the environment around the production area. This cooperation intends to reduce CO_2 and maintain good and healthy air quality in those areas. PT Gunung Raja Paksi Tbk (GRP) officially signed a cooperation agreement with the Sharingyuk Community from Institut Pertanian Bogor (IPB) in September 2021.

The company is committed to support the Government program towards green industry that recognized for ten years in a row by the PROPER certificate awarded by the Republic of Indonesia's Ministry of Environment and Forestry. Aspects of the assessment include environmental permits, water pollution control, air pollution control and management of hazardous and toxic waste.

PT Gunung Raja Paksi Tbk has been obtained ISO 14001:2015 – Environmental Management System certificate. Every 6 months, PT Gunung Raja Paksi Tbk conduct environmental monitoring, such as measuring ground water, air, chimney emission, etc for reporting to UKL-UPL (Upaya Pengelolaan Lingkungan dan Upaya Pemantauan Lingkungan).

PT Gunung Raja Paksi Tbk (GRP) became Indonesia's first and one of Asia's first steel mills to purchase carbon credits. The carbon credits were purchased through Gunung Capital from Climate Impact X (CIX) as much as 10 000 tonnes of carbon credits in October 2021. The carbon credits were purchased from eight recognised Natural Climate Solution (NCS) projects at USD 8.00 per tonne. The NCS projects are global selection of quality projects with spanning reforestation and avoided deforestation initiatives – across Africa, Asia and Latin America – demonstrate the high-performance of carbon sequestration and high levels of verified co-benefits, such as supporting biodiversity, along with social and economic development in local communities.

Version History

1st Version (2022-02-22)

Original version of the EPD

2nd Version (2024-01-24)

The company initiated a thorough review of the original version inventory to ensure accuracy. A miscalculation was identified and promptly rectified. This adjustment was made to provide the most precise and reliable information.

The key differences between this version and the original version of the EPD are as follows:

1. Electricity consumption in the Steel Melting Shop has been revised due to a miscalculation in the previous version electricity metering. Intensity of electricity consumption decrease around 32%.
2. The amount of slab production in the Steel Melting Shop has been updated, resulting in a decrease of approximately 7%. This has led to slight changes in the yield calculations for Module D.
3. The characterization factor for water use has been uniformly applied based on the specific watershed level of the region, replacing the previously used national (Indonesia) average

3rd Version (2025-11-21)

The company has initiated an updated assessment based on the latest data period to ensure up-to-date results. The key differences between this version and previous versions are as follows:

1. The assessment now uses the most recent 2024 production data, resulting to more improved and updated inventory and a more comprehensive inclusion of product grades
2. Use of the updated database and software, i.e., Ecoinvent v3.10 and SimaPro Developer software v10.2.0.1
3. Implementation of the updated PCR 2019:14 Construction Products to align with latest version (v2.0.1).
4. Selection of datasets have been refined to better match operational conditions, improving relevance and reliability.
5. Compared to the previous version, reductions were observed in majority of impact categories, including global warming potential (GWP), ozone depletion potential (ODP), acidification potential (ADP) and more. However, increases were noted in water depletion potential (WDP) attributed to updates in the inventory data and database version.

Abbreviations

Abbreviation	Definition
General Abbreviations	
ABS	American Bureau of Shipping
ACRS	Australasian Certification Reinforcing and Structural Steel
ADP-fossil	Abiotic depletion potential for fossil resources
ADP-minerals & metals	Abiotic depletion potential for non-fossil resources
AGC	Automatic Gauge Control
ANZSIC	Australian and New Zealand Standard Industrial Classification
AP	Acidification potential
AS/NZS	Australian/New Zealand Standard
BKI	Biro Klasifikasi Indonesia
BSI	British Standards Institution)
BV	Bureau Veritas
CEN	European Committee for Standardization
CPC	Central product classification
CRU	Components for reuse
EEE	Exported electrical energy
EET	Exported thermal energy
EF	Environmental Footprint
EN	European Norm (Standard)
EPD	Environmental Product Declaration
EP-freshwater	Eutrophication potential - freshwater

Abbreviation	Definition
General Abbreviations	
EP-marine	Eutrophication potential - marine
EP-terrestrial	Eutrophication potential - terrestrial
ETP-fw	Eco-toxicity - freshwater
FW	Use of net fresh water
GLO	Global
GPI	General Programme Instructions
GRP	Gunung Raja Paksi
GSS	Gunung Steel Group
GWP-biogenic	Global Warming Potential - biogenic
GWP-fossil	Global Warming Potential - fossil fuels
GWP-GHG	Climate Impact
GWP-luluc	Global Warming Potential - land use and land-use change
GWP-total	Global Warming Potential - total
HTP-c	Human toxicity - cancer effects
HTP-nc	Human toxicity - non-cancer effects
HWD	Hazardous waste disposed
ID	Indonesia
IRP	Ionizing radiation - human health

Abbreviation	Definition
General Abbreviations	
ISO	International Organization for Standardization
JIS-JQA	Japanese Industrial Standards (JIS) - Japan Quality Assurance Organization (JQA).
LCA	Life Cycle Assessment
LR	Lloyd's Register
MER	Materials for energy recovery
MFR	Materials for recycling
ND	Not Declared
NHWD	Non-hazardous waste disposed
NRSF	Use of non-renewable secondary fuels
ODP	Depletion potential of the stratospheric ozone layer
PCR	Product Category Rules
PENRE	Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials
PENRM	Use of non-renewable primary energy resources used as raw materials
PENRT	Total use of non-renewable primary energy resources
PERE	Use of renewable primary energy excluding renewable primary energy resources used as raw materials

Abbreviation	Definition
General Abbreviations	
PERM	Use of renewable primary energy resources used as raw materials
PERT	Total use of renewable primary energy resources
PM	Particulate Matter emissions
POCP	Formation potential of tropospheric ozone
PSC	Plate Service Center
PSM	Plate&Steckel Mill
RSF	Use of renewable secondary fuels
RWD	Radioactive waste disposed
SIRIM	Standard and Industrial Research Institute of Malaysia
SM	Use of secondary materials
SMS	Steel Melting Shop
SNI	Indonesian National Standard
SQP	Land use related impacts / soil quality
TS	Tensile Strength
WDP	Water use deprivation potential
YS	Yield Strength

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SOUTHEAST ASIA EPD
INTERNATIONAL EPD SYSTEM

Environmental Product Declaration

GRP
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